

## 1 Assignment 4a — Neutrino Oscillations in Matter



The neutrino time evolution in matter is governed by Schrödinger equation with an effective Hamiltonian that includes both the vacuum term and a matter potential term.

In class, we saw that we can write the time evolution as

$$i \frac{d}{dt} \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \left[ U \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix} U^\dagger + \begin{pmatrix} V_{CC} & 0 \\ 0 & 0 \end{pmatrix} \right] \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix}, \quad (1.1)$$

where  $E_i^0 \simeq p + \frac{m_i^2}{2p}$  are the energy eigenvalues in vacuum,  $U = R(\theta) \equiv \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ , and  $V_{CC}$  is the charged-current matter potential.

- (4 points) 1. Define the effective mixing angle in matter,  $\theta_m$ , such that the effective Hamiltonian can be diagonalized as

$$\hat{H}_m = U_m \begin{pmatrix} E_1^m & 0 \\ 0 & E_2^m \end{pmatrix} U_m^\dagger, \quad \text{where} \quad i \frac{d}{dt} \nu_i^m = E_i^m \nu_i^m, \quad (1.2)$$

where  $U_m = R(\theta_m)$  is the mixing matrix with angle  $\theta_m$  that takes us to the diagonal basis in matter  $\nu_{\text{flavor}} = U_m \nu^m$ . The script  $m$  indicates that these are the matter eigenstates and eigenvalues. We have implicitly assumed that the matter potential is constant throughout the neutrino propagation (take  $t \simeq x$  and Eq. (1.1) above tells you how the state evolves as it propagates from source to detector). Derive an expression for the eigenvalues  $E_i^m$  and  $\sin 2\theta_m$  in terms of the vacuum mixing angle  $\theta$ , the neutrino energy  $E_\nu$ , the mass-squared difference  $\Delta m^2 = m_2^2 - m_1^2$ , and the matter potential  $V_{CC}$ . Define a matter-basis mass splitting as  $\Delta m_m^2 / (2E_\nu) \equiv E_2^m - E_1^m$  and write your expression for  $\Delta m_m^2$  as a function of these parameters.

(Hint: Recall that diagonal terms in the time evolution will not contribute to oscillations.)

(Hint 2: Recall also that a real symmetric  $2 \times 2$  matrix  $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$  can be diagonalized by an orthogonal transformation with angle  $\theta$  such that  $\tan 2\theta = \frac{2c}{b-a}$  or  $\sin 2\theta = \frac{2c}{\sqrt{(b-a)^2 + 4c^2}}$ .

(Hint 3: Your answer should look like  $\sin 2\theta_m = \sin 2\theta / R$  and  $\Delta m_m^2 = \Delta m^2 \sqrt{R}$  for some  $R$  factor.)

- (2 points) 2. Using the expression derived in Question 1, find the condition for which the effective mixing angle in matter  $\theta_m$  becomes maximal (i.e.,  $\theta_m = \pi/4$ ). What does this mean for oscillations? This condition is known as the Mikheyev-Smirnov-Wolfenstein (MSW) resonance condition.

- (2 points) 3. Given that  $V_{CC} = \sqrt{2}G_F N_e$ , where  $G_F$  is the Fermi constant and  $N_e$  is the electron number density in matter, estimate the neutrino energy  $E_\nu$  at which the MSW reso-

nance occurs for solar neutrinos propagating through the Sun

(Hint: take  $N_e \sim 100N_A \text{ cm}^{-3}$ , where  $N_A$  is Avogadro's number. Assume  $\Delta m^2 = 7.5 \times 10^{-5} \text{ eV}^2$  and  $\theta = 33^\circ$ .)

- (2 points)** 4. The survival probability in matter becomes  $P(\nu_e \rightarrow \nu_e) = 1 - \sin^2 2\theta_m \sin^2 \left( \frac{\Delta m_m^2 t}{4E_\nu} \right)$ , analogous to the vacuum oscillation probability. What happens when the MSW resonance condition is exactly satisfied throughout the entire neutrino propagation for  $\Delta m^2 t / 2E_\nu \gg 1$ ? What about when  $\Delta m_M^2 t / 2E_\nu \ll 1$  (expand the sine functions)?

- (Bonus 2 points)** 5. For a changing matter density, the effective mixing angle  $\theta_m$  also changes along the neutrino trajectory. Derive the adiabaticity condition for which a neutrino that starts as a matter eigenstate  $\nu_1^m$  (or  $\nu_2^m$ ) remains in that eigenstate throughout the propagation. What happens when this condition is satisfied?